

Introduction to Machine Learning Problems Unit

3: Multiple Linear Regression

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1. (a) The target variable is a numeric score from 1 to 5.
- (b) We can denote the frequencies as f_{good} , f_{bad} , and $f_{doesn'twork}$. Then, a possible linear model would be:

$$y = \beta_0 + \beta_1 f_{good} + \beta_2 f_{bad} + \beta_3 f_{doesn'twork} + \beta_4 x \quad (1)$$

- (c) We can normalize both scales to values between [0,1].
 - (d) i. $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5$
ii. $\beta_0 + \beta_1 x_{bad} + \beta_2 x_{good}$
iii. If there is no numeric rating, we can check the text for tone based on the word usage in the text.
 - (e) I would use a fraction of reviews with the word "good" because of biasing issues. If two average products had a million reviews and the other had 5, there would be clear biasing issues towards the product with a million reviews.
2. (a) $y = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$
 - (b) We can find $\hat{\beta}$ using

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad (2)$$

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \text{ and } Y = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 7 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}, \text{ and } X^T Y = \begin{bmatrix} 15 \\ 10 \\ 11 \end{bmatrix}$$

$\det(X^T X) = 4(4 - 1) - 2(4 - 2) + 2(2 - 4) = 4$, then we can find $(X^T X)^{-1}$

$$(X^T X)^{-1} = \frac{1}{4} \begin{bmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

Plugging in the values for **Eq.3** results in

$$\hat{\beta} = \frac{1}{4} \begin{bmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix} \begin{bmatrix} 15 \\ 10 \\ 11 \end{bmatrix} = \begin{bmatrix} 0.75 \\ 2.5 \\ 3.5 \end{bmatrix} \quad (3)$$

3. (a)

$$\phi(x) = \begin{bmatrix} x_1 e^{-x_1 - x_2} \\ x_2 e^{-x_1 - x_2} \end{bmatrix}$$

$$\beta = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

(b) i. For $x < 1$,

$$\phi(x) = x$$

$$\beta = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

For $x \geq 1$,

$$\phi(x) = x$$

$$\beta = \begin{bmatrix} a_3 \\ a_4 \end{bmatrix}$$

(c) i.

$$\phi = \begin{bmatrix} e^{-x^2} \\ x_1 e^{-x_2} \end{bmatrix}$$

$$\beta = \begin{bmatrix} e^{a_2} \\ a_1 e^{a_2} \end{bmatrix}$$

$$a_2 = \ln(\beta_1) \quad a_1 = \frac{B_2}{B - 1}$$

4. (a) There are $1 \dots M$ and $0 \dots N$ coefficients, totaling $M + N + 1$ parameters.

(b) We need to know if M or N is larger

$$y = \begin{bmatrix} y_{\max(M,N)} \\ y_{\max(M,N)+1} \\ \vdots \\ y_{T-1} \end{bmatrix} A = \begin{bmatrix} y_{\max(M,N)-1} & \cdots & x_{\max(M,N)} & \cdots & x_{\max(M,N)-N} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ y_{T-1} & \cdots & x_{T-1} & \cdots & x_{T-1-N} \end{bmatrix}$$

5. (a)
$$A = \begin{bmatrix} 1 & \cdots & 1 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \cos(\Omega_1(N-1)) & \cdots & \cos(\Omega_L(N-1)) & \sin(\Omega_1(N-1)) & \cdots & \sin(\Omega_L(N-1)) \end{bmatrix}$$

and
$$\beta = \begin{bmatrix} a_1 \\ \vdots \\ a_L \\ b_1 \\ \vdots \\ b_L \end{bmatrix}$$

(b) If the frequencies were not known, then matrix A would no longer be deterministic and breaks the linearity definition $y = A\beta$.

6. (a)

```
1 X_feat = np.column_stack([X[:,0], X[:,1], X[:,1] * X
2   [:,2]])
2 yhat = X_feat @ beta
```

(b)

```
1 yhat = np.sum(alpha * np.exp(-x[:,None] * beta, axis
= 1))
```

(c)

```
1 dist = np.sum((x[:,None,:] - y[None,:m:]) **2, axis
= 2)
```